Modelling of heat transfer and vaporization in channel-based VLM thrusters

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1 Introduction

This work documents and analyzes a one-dimensional thermal–fluid model for a Vaporizing Liquid Microthruster (VLM). A resistive heater supplies heat to a working fluid (water) flowing through serpentine microchannels; the fluid is heated, vaporized, and then expanded through a planar micro-nozzle to produce thrust. The model marches axially along a representative serpentine path (replicated across N_{channels} in parallel) and predicts the evolution of its temperature $T(x)$, pressure $p(x)$, phase quality $x(x)$, and void fraction $\alpha(x)$. Heat transfer from the heater to the fluid is mediated by a multilayer thermal stack (heater/TBR/SiN/Si and convection), and the unknown wall temperature T_w is obtained from a steady energy balance including backside conduction and radiation. At the channel exit, nozzle performance is estimated using isentropic relations corrected for discharge losses and axial projection by the divergence half-angle θ.

Within this scope, the report addresses two research questions:

Research questions

1. What is a suitable set of equations and assumptions for modelling the heat transfer and vaporization in the given VLM geometries?
2. How does the heating and vaporization actually work in the given VLM geometries, and which values of the physical parameters in the thruster can be expected under given operational conditions?

We answer the first by laying out the constitutive equations, coupling assumptions, and simplifications (laminar regime, homogeneous-equilibrium two-phase treatment, curvature-enhanced heat transfer, and a wall energy balance). We answer the second by applying the model to representative operating points and reporting the resulting axial fields, nozzle performance, and sensitivity to geometry and power.

2 Literature review and modeling provenance

This section describes the papers used for this assignment.

Geometry reconstruction from Silva et al.

This paper presents the design, manufacturing, and characterization of Vaporizing Liquid Microthrusters (VLMs) developed at TU Delft for use in small satellites like CubeSats and PocketQubes. The devices are built using silicon-based MEMS technology and use water as the propellant. A key feature is their integrated molybdenum heaters, which serve a dual purpose: they vaporize the liquid propellant and simultaneously function as temperature sensors, as the heater's electrical resistance is used to estimate the temperature in the vaporizing chamber. The authors describe the complete manufacturing process and the results of characterization tests, ultimately demonstrating the successful operation of the thrusters and validating the integrated temperature sensing method.

The overall serpentine channel length used in the solver was reconstructed from Fig. 2 of Silva *et al.* [1]. The image shows repeated straight and curved sections; the total length L was obtained by counting these segments and multiplying by the corresponding section lengths read from the figure.

Key takeaways from Silva et al. [1]

- Serpentine microchannel layout and dimensions (throat widths, divergent lengths) are accessible from annotated figures; when tabular data is missing, careful measurement from figures yields usable first-order geometry.
- Planar nozzle geometry (throat and exit widths, divergent length) enables estimating the discharge angle θ via simple trigonometry, suitable for momentum projection in thrust calculations.

Heat transfer and friction in curved microchannels (Rosgueti et al.)

The solver's Nusselt number and friction-factor enhancements for serpentine (curved) channels follow the spirit of Rosgueti *et al.* [2], who report Dean-number-driven augmentation of heat transfer and pressure drop in curved microchannels. In this work we implement:

- An enhancement factor to calculate friction in serpentine micro-channels.
- An enhancement factor to calculate Nusselt number in serpentine micro-channels.

On CHF in serpentine microchannels

No literature was available for critical heat flux (CHF) data and correlations applicable to serpentine/curved microchannels and planar electrothermal thruster geometries comparable to those modeled here. No directly transferable CHF models or validated data sets were found for the specific combination of length scales, materials, and two-phase operating conditions considered. Consequently, CHF prediction is omitted in this version of the model. Instead, the formulation uses a conservative two-phase treatment.

3 Constants and geometry

Provenance of values

Unless otherwise noted, all geometric quantities reported in this section (serpentine length, heater footprint areas, wall thicknesses, outer diameters, and planar nozzle dimensions including throat/exit widths, divergent lengths and depth) are extracted directly from Fig. 2 and associated descriptions in Silva *et al.* [1]. Non-geometric thermophysical constants (material properties, environmental conditions) follow standard reference values or in-code empirical fits.

Operating conditions

Quantity	Value
Applied voltage	$V_{\text{applied}} = 5\,\text{V}$
Total inlet mass flow	$\dot{m}_{\text{total}} = 0.82 \cdot 10^{-6}\,\text{kg}\,\text{s}^{-1}$
Inlet pressure	$p_{\text{in}} = 5.0 \cdot 10^{5}\,\text{Pa}$
Inlet temperature	$T_{\text{in}} = 293\,\text{K}$

Numerical discretization

Parameter	Value
Axial slices	$N_{\text{slices}} = 200$
Parallel channels	$N_{\text{channels}} = 5$

Geometric parameters (from Silva et al.)

Item	Symbol	Large case	Small case
Heater footprint area	A_H	$5.4 \cdot 10^{-6}\,\text{m}^2$	$5.16 \cdot 10^{-6}\,\text{m}^2$
Wall thickness	t_{wall}	54 µm	20 µm
Outer diameter	D_o	266 µm	60 µm
Inner diameter	$D_i = D_o - 2t_{\text{wall}}$	derived	derived
Hydraulic diameter	$D_h = D_i$	derived	derived
Channel length	L	8.96 mm	
Planar depth (nozzle)	d	100 µm	

Planar nozzle dimensions (from Silva et al.)

Widths and divergent lengths (all in µm); discharge coefficients C_d assigned per type.

Type	w_t	w_{nd}	l_{nd}	C_d	Exit angle θ	Notes
L	45	500	645	0.92	inferred	Longer divergent section
W	45	780	660	0.88	inferred	Wider, larger angle
B	45	500	500	0.90	inferred	Short divergent length

Material and thermal properties

Silicon thermal conductivity uses a temperature-dependent fit: $k_{Si}(T) = 1.34122{\times}10^5 T^{-1.2073}\,\text{W}\,\text{m}^{-1}\,\text{K}^{-1}$, following [3]. Molybdenum properties (thermal/electrical) are obtained from [4]. Representative values:

Parameter	Value/Expression
Silicon nitride conductivity	$k_{SiN} \approx 3\,\text{W}\,\text{m}^{-1}\text{K}^{-1}$
SiN thickness	$t_{SiN} = 500\,\text{nm}$
Silicon conduction thickness	$t_{Si} = 100\,\mu\text{m}$
TBR (Mo/SiN)	$R''_{\text{Mo/SiN}} \approx 2 \cdot 10^{-8}\,\text{m}^2\text{KW}^{-1}$
TBR (SiN/Si)	$R''_{\text{SiN/Si}} \approx 2 \cdot 10^{-8}\,\text{m}^2\text{KW}^{-1}$
Heater resistances	$R_1 = 3.40\,\Omega,\ R_2 = 2.38\,\Omega$

Environmental and loss parameters

Quantity	Value
Ambient/base temperature	$T_\infty = T_{base} = 293.15\,\text{K}$
Surface emissivity	$\varepsilon = 0.8$

Derived relations

Inner diameter follows $D_i = D_o - 2t_{\text{wall}}$; hydraulic diameter $D_h = D_i$. Heater power for a selected heater is $P = V^2/R$.

4 Boundary conditions and main assumptions

Summary

Steady, one-dimensional axial march from channel inlet to nozzle entrance. Inlet state is imposed; flow assumed laminar throughout. Two-phase thermodynamics use a homogeneous equilibrium (HEM) mixture with instantaneous saturation at boiling.

Mathematical statements

Inlet state

$$T(0) = T_{\text{in}}, \qquad p(0) = p_{\text{in}}, \qquad \dot{m}_c = \frac{\dot{m}_{\text{total}}}{N_{\text{channels}}}.$$

Domain and regime

Steady, 1D axial formulation; laminar flow assumed (validated by resulting $\text{Re} < 2300$). No turbulent or transitional correlations are invoked.

Two-phase model

Homogeneous equilibrium mixture (HEM) with instantaneous saturation in boiling:

$$\rho_m^{-1} = \frac{x}{\rho_v} + \frac{1-x}{\rho_l}, \qquad T = \begin{cases} T, & x = 0, \\ T_{sat}(p), & 0 < x < 1, \\ T, & x = 1. \end{cases}$$

What HEM assumes

Homogeneous Equilibrium Model (HEM) assumes: (i) mechanical equilibrium (common pressure), (ii) thermal equilibrium (common temperature; in boiling the mixture is at $T_{sat}(p)$), and (iii) kinematic homogeneity. Mixture density follows $\rho_m^{-1} = x/\rho_v + (1-x)/\rho_l$; other transport properties use mixing rules.

5 Channel hydraulics and heat transfer

5.1 Flow and friction (laminar regime)

For a single channel: mean velocity $u = \dot{m}_c/(\rho A)$, Reynolds number $\text{Re} = \rho u D_h/\mu$. Across all simulated cases, the flow remains laminar ($\text{Re} < 2300$), so we use the laminar smooth-pipe relation

$$f_s = \frac{64}{\text{Re}}. \tag{1}$$

The serpentine friction enhancement e_f is used. The axial pressure gradient is

$$\frac{\mathrm{d}p}{\mathrm{d}x} = -\frac{f\,\rho u^2}{2D_h}, \qquad f = e_f\, f_s\, \phi_f^2, \tag{2}$$

where $\phi_f^2 = 1 + 10x(1-x)$ is the two-phase friction multiplier.

5.2 Nusselt number with curvature and development

Define Dean number $\mathrm{De} = \mathrm{Re}\sqrt{D_h/(2R_c)}$. In laminar flow with constant wall heat flux, the baseline is $\mathrm{Nu}_\ell = 4.36$. Curvature enhances mixing; we apply a conservative Dean multiplier consistent with curved-microchannel trends (Rosgueti *et al.* [2]):

$$e_{De} = 1 + 0.10\,\mathrm{De}^{0.5}, \quad 1 \le e_{De} \le 3 \tag{3}$$

The effective Nusselt number is then (with serpentine enhancement factor e_{Nu} from Rosgueti *et al.* [2])

$$\mathrm{Nu} = e_{Nu}\,\underbrace{\mathrm{Nu}_\ell}_{4.36}\,e_{De}\,\eta_{\mathrm{dev}}, \qquad 3 \le \mathrm{Nu} \le 200, \qquad h = \frac{\mathrm{Nu}\,k}{D_h}. \tag{4}$$

5.3 Heater-to-wall conduction stack

From the resistive heater to the fluid, heat traverses a series of layers and interfaces modeled as resistances per unit area. With temperature-dependent silicon conductivity $k_{Si}(T)$, the net series resistance is

$$R''_{\mathrm{stack}} = R''_{\mathrm{Mo/SiN}} + \frac{t_{SiN}}{k_{SiN}} + R''_{\mathrm{SiN/Si}} + \frac{t_{Si}}{k_{Si}(T)} + R''_{\mathrm{TCR}} + R_{\mathrm{conv}}, \tag{5}$$

where $R_{\mathrm{conv}} = 1/\mathcal{H}_c$ is the convective term defined below and R''_{TCR} accounts for inefficiencies. The two thin-film interface resistances (Mo/SiN and SiN/Si) capture thermal boundary resistance (TBR) effects.

Conduction path: heater $\rightarrow$ fluid
Mo/SiN (TBR) $\rightarrow$ SiN film $\rightarrow$ SiN/Si (TBR) $\rightarrow$ bulk Si ($k_{Si}(T)$) $\rightarrow$ convection to fluid.

5.4 Wall-to-fluid coupling and wall energy balance

Per-heater-area convective conductance uses the total wetted perimeter of all channels normalized by heater footprint per unit length $a' = A_H/L$:

$$\mathcal{H}_c = h\,\frac{P_{w,\mathrm{tot}}}{a'}, \qquad R_{\mathrm{conv}} = \frac{1}{\mathcal{H}_c}. \tag{6}$$

Using R''_{stack} defined above, the wall temperature T_w solves a per-area steady energy balance

$$q''_{\mathrm{elec}} = \underbrace{\frac{T_w - T_f}{R''_{\mathrm{stack}}}}_{q''_f} + \underbrace{\varepsilon\sigma\,(T_w^4 - T_\infty^4)}_{q''_{rad}} + \underbrace{h_{ext}(T_w - T_\infty)}_{q''_{ext}}. \tag{7}$$

6 Nozzle geometry and isentropic model with losses

6.1 Planar geometry and discharge angle

The nozzle is approximated as a rectangular cross-section of depth d. Throat and exit areas are $A_t = w_t\,d$ and $A_e = w_{nd}\,d$. The planar divergence half-angle is

$$\theta = \arctan\left(\frac{w_{nd} - w_t}{2\,l_{nd}}\right), \tag{8}$$

which causes an axial thrust projection factor $\cos\theta$.

6.2 Isentropic relations and discharge coefficient

Assuming ideal-gas isentropic expansion with ratio of specific heats γ and gas constant R, the area–Mach relation gives the supersonic exit Mach number M_e from A_e/A_t. The choked (sonic) mass flow at the throat, including a discharge coefficient $C_d \in (0,1)$, is

$$\dot{m}_* = C_d \, \frac{A_t \, p_0 \, \gamma \sqrt{\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{\gamma-1}}}}{\sqrt{\gamma \, R \, T_0}}. \tag{9}$$

Given M_e, exit temperature and pressure are

$$T_e = \frac{T_0}{1 + \frac{\gamma-1}{2} M_e^2}, \qquad p_e = p_0 \left(1 + \frac{\gamma-1}{2} M_e^2\right)^{-\frac{\gamma}{\gamma-1}}, \tag{10}$$

with exhaust speed $V_e = M_e \sqrt{\gamma R T_e}$. The model limits the nozzle mass flow to the smaller of the choked capacity and the available vapor mass flow $\dot{m}_v = x_{\text{exit}} \, \dot{m}_{\text{total}}$. The thrust and specific impulse (projected axially) are

$$F = \dot{m} \, V_e \cos\theta + (p_e - p_a) A_e, \qquad I_{sp} = \frac{F}{\dot{m} \, g_0}. \tag{11}$$

Lower C_d captures viscous, non-ideal, and end effects; a larger θ reduces the axial momentum component via $\cos\theta$. Nozzle widths and divergent lengths used to infer θ are taken from Silva *et al.* [1].

7 Results overview

Table 1 lists the 12 modeled cases across heaters (H1, H2), geometries (Large/Small), and nozzle types (L, W, B). The final values for thrust and specific impulse are feasible.

Table 1: Compact results summary (selected columns).

Case	Ht	Geom	Noz	P [W]	Q [W]	Util	$\dot{m}_N$ [mg/s]	F [mN]	I_{sp} [s]
1	H1	Large	L	7.4	2.19	0.30	0.78	0.96	126.0
2	H1	Large	W	7.4	2.19	0.30	0.78	0.89	117.2
3	H1	Large	B	7.4	2.19	0.30	0.78	0.93	122.4
4	H1	Small	L	7.4	2.62	0.36	0.28	0.33	121.3
5	H1	Small	W	7.4	2.62	0.36	0.27	0.30	115.1
6	H1	Small	B	7.4	2.62	0.36	0.27	0.31	117.4
7	H2	Large	L	10.5	2.73	0.26	0.82	1.22	151.2
8	H2	Large	W	10.5	2.73	0.26	0.82	1.14	141.4
9	H2	Large	B	10.5	2.73	0.26	0.82	1.18	146.8
10	H2	Small	L	10.5	2.90	0.28	0.24	0.33	138.6
11	H2	Small	W	10.5	2.90	0.28	0.23	0.30	131.6
12	H2	Small	B	10.5	2.90	0.28	0.24	0.31	134.2

Pressure and temperature distributions along the heating chamber.

The plots show axial profiles of temperature, pressure, and quality.

Case 1 (H1, Large geometry). Temperature rises from the inlet until the local saturation temperature is reached. Upon the onset of boiling, the fluid temperature pins to $T_{sat}(p)$ while added heat goes into latent energy. The profile stays at saturation nearly to the exit because not all of the liquid is vaporized; the predicted exit quality is $x \approx 0.945$. A subtle kink in the pressure trace appears when it starts boiling. Physically, the onset of vapor formation reduces the mixture density and increases velocity, and the two-phase friction multiplier $\phi_f^2 = 1 + 10x(1-x)$ begins to grow from unity. Both effects raise $|\mathrm{d}p/\mathrm{d}x|$, so the pressure drops slightly faster immediately after boiling begins.

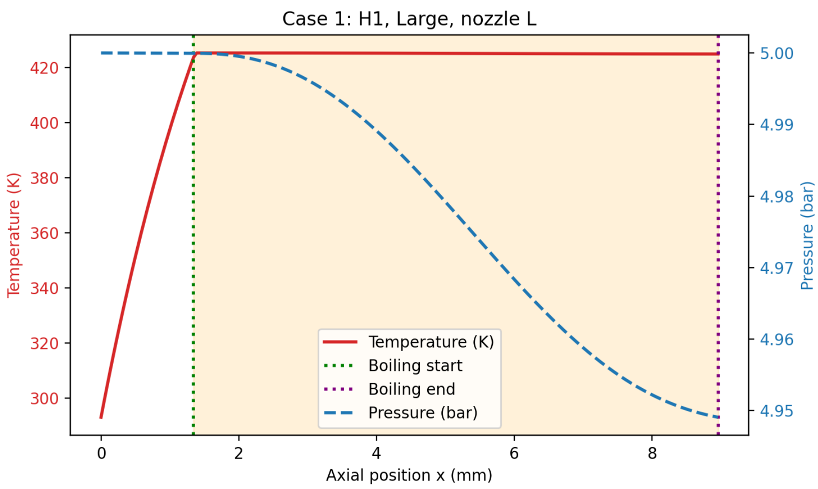

Figure 1: Case 1 (H1, Large geometry). Temperature rises to saturation and remains at $T_{sat}(p)$ while boiling proceeds; exit quality is $x \approx 0.945$. A mild increase in pressure gradient appears at boiling onset due to decreasing density and increasing two-phase friction.

Case 4 (H1, Small geometry). The temperature increases to $T_{sat}(p)$, after which a pronounced pressure drop occurs. This steep decline is much stronger than in Case 1 because the small geometry has a much smaller hydraulic diameter D_h, yielding larger friction losses via $\mathrm{d}p/\mathrm{d}x \propto f\,\rho u^2/(2D_h)$. With the same per-channel mass flow on a smaller cross-section, velocity and dynamic pressure are higher, and once boiling begins the rapid density reduction further accelerates the flow. As pressure plummets, the saturation temperature drops with p, so the fluid temperature also decreases while boiling continues. When the liquid fully vaporizes ($x \to 1$), sensible heating resumes and the vapor temperature rises again toward the exit.

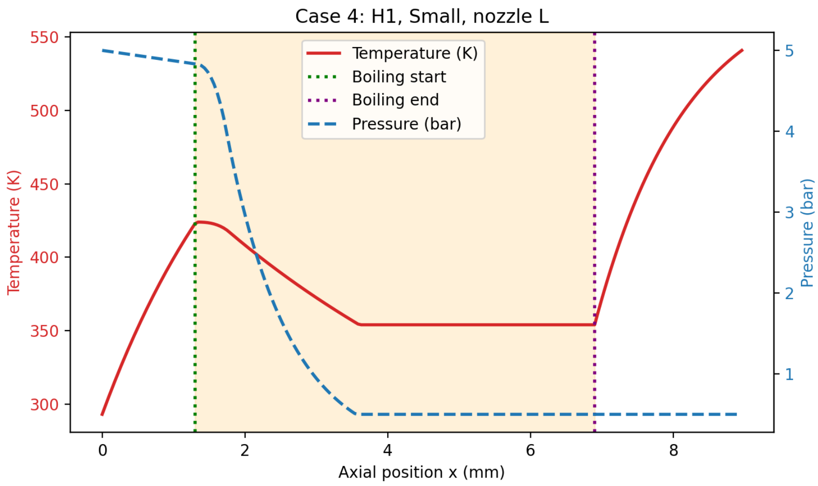

Figure 2: Case 4 (H1, Small geometry). A pronounced two-phase pressure drop develops after boiling onset because the smaller hydraulic diameter raises friction losses and velocity; $T_{sat}(p)$ falls with pressure, producing a temporary temperature decrease until complete vaporization.

Case 7 (H2, Large geometry). This case mirrors the qualitative behavior of Case 1 but with higher electrical power. The added power vaporizes the flow well before the exit, leaving a longer downstream length for superheating the vapor. Consequently, exit vapor temperature and thrust increase relative to Case 1.

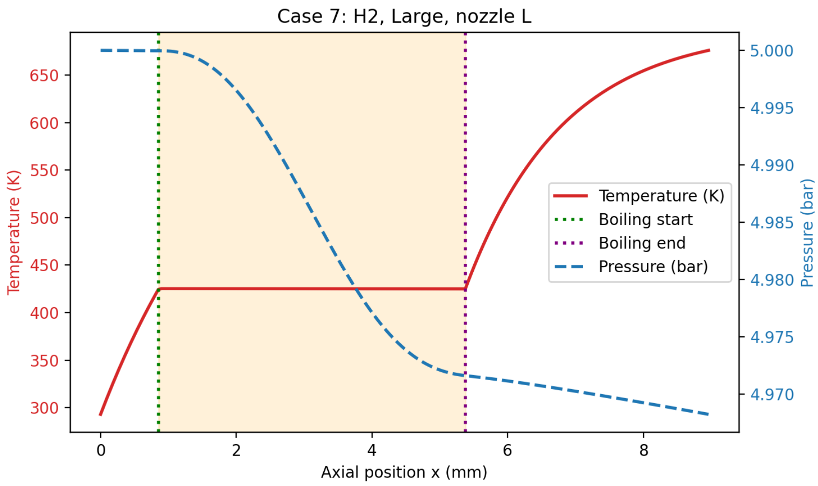

Figure 3: Case 7 (H2, Large geometry). Higher power drives complete boiling well upstream of the exit, leaving more length to superheat the vapor.

Case 10 (H2, Small geometry). The evolution resembles Case 4 (same small geometry), including the strong depressurization through the boiling region. The higher power drives the system to a higher final temperature after complete vaporization, but the unrealistic magnitude of the pressure drop persists.

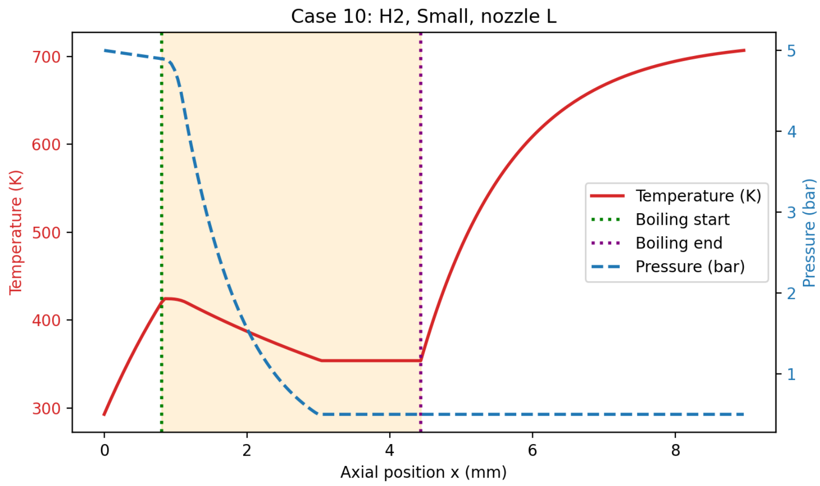

Figure 4: Case 10 (H2, Small geometry). Similar to Case 4 with strong depressurization through boiling; higher power yields a higher final vapor temperature by the exit.

Interpretation and model limitation

For the small-geometry cases, the steep depressurization indicates that the present friction model likely overpredicts two-phase pressure losses at these scales. It performs reasonably for the larger geometry but appears to "break" for the smaller one, leading to exit pressures that are not physically plausible and to the non-monotone temperature behavior tied to $T_{sat}(p)$. Addressing this requires a microchannel-specific two-phase pressure-drop correlation.

8 Conclusions

Answer to the research questions

RQ1. A concise, traceable model for these VLM geometries is a 1D laminar axial march with Dean-number curvature enhancements for friction and heat transfer, a homogeneous equilibrium (HEM) two-phase treatment (temperature locked to saturation during boiling), a multi-layer wall conduction balance (heater/TBR/SiN/Si + radiation + external convection), and an isentropic planar nozzle with a discharge coefficient and axial projection. This set captures the main coupled phenomena (heating, boiling onset, latent absorption, superheating) with limited computational cost.

RQ2. Large channels show moderate pressure decline and a stable saturation plateau; increasing power moves complete vaporization upstream and allows downstream superheating (raising thrust and I_{sp}). Small channels display very steep pressure drops during boiling and a temporary temperature dip as falling pressure lowers T_{sat}; this likely reflects friction overprediction, not true device behavior. Representative thrust lies around $0.9{-}1.2\,\mathrm{mN}$ for large geometry and somewhat lower for small; specific impulse reaches $\sim 150\,\mathrm{s}$ at higher power.

Key observations

- Higher heater power (H2) shifts complete vaporization upstream, enabling vapor superheat and higher performance.
- Small geometry predictions show unrealistically deep depressurization, flagging the need for better two-phase friction correlations.

Main limitations and next steps

- HEM and generic friction multiplier likely overestimate two-phase pressure drop in tight channels.
- CHF not modeled; high-flux stability margins remain unknown.
- Curvature/boiling heat-transfer enhancements are not dimension-specific correlations.
- Nozzle model ignores detailed viscous and non-equilibrium effects.

References

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